DE–710 **101**

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, DECEMBER 2014.

First Semester

DIGITAL COMPUTER ORGANISATION

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Convert the following :

   (a) $(F2EC)_{16} = (?)_2$

   (b) $(6745)_{10} = (?)_{16}$

   (c) $(SA4)_{16} = (?)_8$

   (d) $(8364)_{10} = (?)_{16}$.

2. State and prove Demorgan's theorem.

3. What is demultiplexer? Discuss the function of a demultiplexer.

4. Obtain the simplified expression in SUM of PRODUCT form.

   (a) $F(A,B,C) = \sum(3,4,5,6,7)$

   (b) $F(A,B,C,D) = \sum(4,6,7,15)$.

5. What is the function A a shift register? What are its various types? Explain.

6. What are NOR and NAND gates? Why are they called universal gates? Explain.

7. Distinguish between the static RAM and Dynamic RAM.

8. List the functions of I/O interface.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. What is full-adder? How is a full-adder built using half-adders? Write the truth-table for a full-adder and explain.

10. Draw the schematic diagram of a master-slave J-k flip-flop. Discuss its working principles. What are its advantages over other types of flip-flop?

11. Describe in detail about the fetch and execution cycle with suitable diagrams.

12. What is a counter? Design a Mod-5 counter with any one of the flip-flop type R-s/J-k.

13. (a) Explain briefly about maskable interrupt.

    (b) Explain D flip-flop with necessary diagram.

14. Explain the following in detail :

    (a) Static memory

    (b) Dynamic memory.

15. Explain the DMA data transfer with a schematic diagram.

DE–711 **102**

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, DECEMBER 2014.

First Semester

C AND DATA STRUCTURES

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Write a note on break, continue and goto statements.

2. List any eight library functions available in C.

3. Write a program in C to find prime numbers between 1 to 100.

4. How do you pass an array to a function? Explain.

5. Write a program in C to create a text file in C.

6. Define queue. Explain the circular queue.

7. Briefly discuss the tree data structure.

8. Define binary tree. Explain the various binary tree traversal methods.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. What are the types of control statements available in C? Explain with example.

10. Discuss the various types of arrays available in C.

11. Write a note on :

    (a) The pre-processor

    (b) Passing pointers to function

    (c) Dynamic memory allocation.

12. Write a program in C to perform read and append operation in a text file.

13. List the applications of stack data structure.

14. Define linked list. Explain the various types of linked list.

15. Write a program in C to print the Armstrong numbers using pointers and functions.

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DE–712 **103**

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, DECEMBER 2014.

First Semester

RELATIONAL DATABASE MANAGEMENT SYSTEMS

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following questions.

1. Discuss the advantages of RDBMS over traditional file management system.

2. Define relational model and its advantages.

3. List and explain the important rules of subqueries.

4. Write the syntax for creating a table, inserting record into the table and deleting the records from the table.

5. Explain the operators in relational algebra through examples.

6. Differentiate between SQL and PL/SQL.

7. What are the advantages and disadvantages of triggers?

8. Define view and its advantages over table.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR of the following questions.

9. Discuss the various data types of SQL.

10. Discuss in detail the embedded SQL statement.

11. Describe the three levels of data abstraction.

12. Explain the structure of PL/SQL program. Explain the different parameter modes for subprograms.

13. Explain DML commands in SQL. Explain the various joints in SQL.

14. Explain the following commands.

    (a) Commit

    (b) Rollback

    (c) Save point.

15. Describe the basic features of SQL statement. List its advantages and disadvantages.

DE–713 **104**

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, DECEMBER 2014.

First Semester

OPERATING SYSTEMS

(2010 academic year onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Explain the dual mode operation.

2. Distinguish between the long term scheduler and short term scheduler.

3. Write a short note on

   (a) Race condition

   (b) Mutual exclusion

   (c) Spin Lock

   (d) Semaphores.

4. Discuss in detail the Interrupt handling techniques.

5. Explain the memory allocation techniques in detail.

6. Discuss about virtual memory using segmentation.

7. How do you handle the memory without swapping (or) paging?

8. What are the security issues in file management? Explain.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Briefly explain the important services of an operating system.

10. Explain the following classical synchronization problems.

    (a) Bounded buffer problem

    (b) Dining philosopher's problem.

11. Discuss in detail the Inter Process Communication.

12. Briefly explain the deadlock avoidance using Banker's algorithm.

13. Describe the page replacement algorithm with an example.

14. Explain the free space management techniques in detail.

15. Suppose that a disk drive has 5,000 cylinders, numbered 0 to 4999. The drive is currently serving a request at cylinder 143, and the privious request was at cylinder l25. The queue of pending requests, in FIFO order, is:

    86, 1470, 913, 1774, 948, 1509, 1022, 1750, 130.

    Starting from the current head position, what is the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests for each of the following disk-scheduling algorithms?

    (a) FCFS

    (b) SCAN

    (c) LOOK.

DE–714 **201**

DISTANCE EDUCATION

M.C.A. (S) DEGREE EXAMINATION, DECEMBER 2014.

Second Semester

OBJECT ORIENTED PROGRAMMING AND C++

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. List the advantages of object oriented programming.

2. Write a C++ program to print the prime numbers from 1 to 100.

3. Describe break, continue and goto statements in C++.

4. Write a C++ program to initialize, sort and print the content of string array.

5. Briefly discuss the inline function in C++.

6. Write a short note on virtual function in C++ with example.

7. Differentiate constructor and destructor.

8. How do you check the end of file operation in C++? Explain.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Write a C++ program to print the sum of squares from 1 to 100.

10. Discuss the various iterative statements available in C++ with example.

11. Explain the polymorphism concept with example.

12. Define inheritance. Explain the various types of inheritance available in C++.

13. List and explain the various functions available in C++ to read and write a content from / to a file.

14. How do you pass an object as an argument of a function in C++? Explain.

15. Write a note on:

    (a) Recursion.

    (b) Macros.

    (c) Copy constructor.

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DE–715 **202**

DISTANCE EDUCATION

M.C.A. (S) DEGREE EXAMINATION, DECEMBER 2014.

Second Semester

UNIX AND SHELL PROGRAMMING

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. How to select the group of filenames? With suitable examples.
2. Write short note on the stream editor sed.
3. Define : Meta characters. Illustrate.
4. What is the use of tee and exit command?
5. Write notes on the formulating of text file and documents in Unix.
6. Write notes on debugging tools.
7. List any two standard I/O functions and explain.
8. Discuss relational operators with examples.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the directories and path names.

10. Write about grep family of commands with options.

11. Explain about input and output redirection with an example.

12. Explain the looping features available in Unix shell programming.

13. Explain the shell process in Unix operating system.

14. Explain the features of Unix.

15. Derive a script menu sh which accepts value from 1 to 5 and performs some action depending on the number keyed in.

DE–716 **203**

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, DECEMBER 2014.

Second Semester

COMMUNICATION SKILLS

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Discuss the importance of communication.

2. Explain the process of communication.

3. What are the expressing techniques?

4. How could a discussion be controlled?

5. What are the participation techniques?

6. Write a short note on interpersonal behaviour.

7. What do you mean by mock – negotiation?

8. Write a short note on personal style.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the techniques involved in the writing articles.

10. How do you distinguish between linear reading and faster reading?

11. How do you plan for an interview?

12. Examine the tool for personality identification.

13. Write an essay on leading and participating meetings.

14. What are the various communication skills for meetings?

15. How do you distinguish between downward and upward communication?

DE–717 **204**

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, DECEMBER 2014.

Second Semester

COMPUTER GRAPHICS

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following questions.

1. What is computer graphics? Explain the usage of computer graphics.

2. Write an algorithm for drawing a circle.

3. Write and explain DDA line drawing algorithm.

4. Explain the homogeneous matrix representation of 2D transformations.

5. Explain the method of representing a 3D object.

6. Write down the transformation matrix for 3D translation.

7. Briefly explain the Cohen Sutherland algorithm for line clipping.

8. Explain the menu design for graphics systems.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR of the following questions.

9. Explain the various interactive input devices in detail.

10. Write a note on various applications of computer graphics.

11. Explain in detail about the ray tracing methods.

12. Explain the hidden line and hidden surface elimination methods.

13. How translation and rotation can be performed on a three dimensional object?

14. Explain in detail about the input and output handling in windows system.

15. Explain the following algorithms

    (a) Z-buffer algorithm

    (b) Bresenham's algorithm.

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DE–718 **301**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2014.

Third Semester

DISCRETE MATHEMATICS

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Show that $((P \vee Q) \wedge \neg(\neg P \wedge (\neg Q \vee \neg R))) \vee (\neg P \wedge \neg Q) \vee (\neg P \wedge \neg R)$ is a tautology.

2. Show that $S \vee R$ is tautologically implied by $(P \vee Q) \wedge (P \to R) \wedge (Q \to S)$.

3. Define the following with example.

   (a) Equal sets

   (b) Proper subset

   (c) Universal set.

4. If $X = \{1,2,3,4\}$ and $R = \{\langle x, y\rangle / x > y\}$. Draw the graph of R and also give its matrix.

5. Write a note on hashing functions with an example.

6. Define subsemigroup and submonoid with an example.

7. Prove that the ring of even integers is a subring of the ring of integers.

8. Define the following with an example.

   (a) Directed graph

   (b) Adjacency matrix of a graph.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Obtain the principal conjunctive normal form of the formula S given by $(\neg P \to R) \wedge (Q \rightleftarrows P)$.

10. Explain statement function, variables and quantifiers with example.

11. (a) Give the relation matrices M_R and M_S, find M_{ROS}, $M_{\tilde{R}}$, $M_{\tilde{S}}$, $M_{\widetilde{ROS}}$.

    (b) Show that, there are only five distinct Hasse diagrams for partially ordered sets that contain three elements.

12. Explain binary operation and n-ary operation with examples.

13. If $\langle S, * \rangle$ be a given semigroup. Then prove that, there exists a homomorphism $g : S \to S^S$, where $\langle S^S, \circ \rangle$ is a semigroup of functions from S to S under the operation of (left) composition.

14. State and prove Lagrange's theorem.

15. Discuss in detail about matrix representation of graphs.

DE–719 **302**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2014.

Third Semester

COMPUTER NETWORKS

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Write a note on
   (a) Connectionless services.
   (b) Services primitives.

2. Give a brief note on communication satellites.

3. Discuss design issues of data – link layer.

4. Write a note on protocol verification.

5. Explain Tunneling and fragmentation with examples.

6. Discuss transport service primitives and berkeley sockets.

7. Discuss UDP in detail.

8. Write short notes on electronic – mail.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain congestion control algorithm.

10. Discuss the interior gateway routing protocol in detail.

11. Explain 'AES' – algorithm with an example.

12. Explain OSI reference model with neat diagram.

13. Explain sliding window protocol with an example.

14. Write a note on
    (a) Distance vector routing algorithm
    (b) Link – state routing algorithm.

15. Discuss SNMP architecture and its components in detail.

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DE–721 **304**

DISTANCE EDUCATION

M.C.A. /M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2014.

Third Semester

VISUAL PROGRAMMING

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Explain the uses of dialog boxes.
2. Write a note on the fundamentals of visual basic
3. Briefly explain the file operations with an example.
4. Explain the various types of modules.
5. Write a short note on event handling.
6. Define MFC? Explain briefly.
7. Discuss the object linking and embedding.
8. Briefly explain the user interface classes.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain in detail the procedure and function with suitable example.

10. Define database? Write down the steps to create an database in visual programming.

11. Explain in detail the document view architecture.

12. Discuss the MFC file handling in detail.

13. Define MDI? How to create an MDI form? Give an example.

14. Explain the mouse and keyboard events.

15. What is ODBC? Explain how to add ODBC capability to an MFC application.

DE–722 **401**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2014.

Fourth Semester

ACCOUNTING AND FINANCIAL MANAGEMENT

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Explain the nature, scope and objectives of Financial Management.

2. Journalise the following transactions of M/S. Ram & Co. 2011 Jan. 1.

Jan 1	Business started with Rs. 3,00,000 and cash deposited with bank 1,00,000
Jan 4	Purchased Machine on credit from Ezhil Rs. 75,000
Jan 10	Goods sold to Ramesh Rs. 30,000
Jan 22	Cash paid Ganesh Rs. 40,000
Jan 24	Paid advertisement expenses Rs. 15,000

3. Discuss the managerial uses of Cash Flow statement.

4. What are the advantages of break-even point?

5. Explain the various sources of Longterm Finance.

6. From the following details find out

(a) Profit Volume Ratio

(b) B.E.P.

(c) Margin of safety

	Rs.
Sales	1,00,000
Total Costs	80,000
Fixed Costs	20,000
Net Profit	20,000

7. The expenses budgeted for production at 10,000 units in a factory are given below :

	Rs. per unit
Materials	70
Labour	25
Variable overheads	20
Fixed overheads (1,00,000)	10
Variable overheads (Direct)	5
Selling Expenses (10% fixed)	13
Administration Expenses (Rs. 50,000)	5
Distribution Expenses (20% Fixed)	7
	155

Prepare a budget of the production.

(a) 8000 units

(b) 6000 units. Assume that the administration expenses are rigid for levels of production.

8. Calculate the present value of the following cash in flow :

(a) Rs. 20,000 to be received after 7 years from now. Assume 8% time preference for money.

(b) Rs. 30,000 to be received after 4 years from now. Assume 15% time preference money.

(c) Rs. 40,000 to be received after 2 years from now. Assume 17% time preference for money.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. On 31st March 2010, the following Trial Balance was extracted from the Books of "RAJ" find out the Net Profit to be transfered to his capital account.

	Debit	Credit
Capital A/C	–	50,000
Plant and Machinery	80,000	–
Sales	–	1,77,000
Purchases	60,000	–
Purchases Returns	1,000	750
Opening Stock	30,000	800
Discount	350	–
Bank Charges	75	–
Sundry Debtors	45,000	25,000
Sundry Creditors	–	–
Salaries	6,800	–
Wages	10,000	–
Carriage Inwards	750	–

	Debit	Credit
Carriage Outwards	1,200	–
Bad Debts Provisions	–	525
Rent, Rates and Taxes	10,000	–
Advertisement	2,000	–
Cash in Hand	900	–
Cash at Bank	6,000	–
	2,54,075	2,54,075

The following adjustment are required :

(a) Closing stock Rs. 35,000

(b) Depreciation on plant 6% p.a.

(c) Bad debts provision to be adjusted to Rs. 500

(d) Interest on capital to be allowed at 5% p.a.

10. The following are the Sumarised balance sheet of X Limited as on 31st December 2010–2011.

Liabilities	2010	2011	Assets	2010	2011
Share Capital	1,00,000	1,00,000	Goodwill	12,000	12,000
General Reserve	14,000	18,000	Buildings	40,000	36,000
Profits and Loss	16,000	13,000	Plant	37,000	36,000
Creditors	8,000	5,400	Investment	10,000	11,000
Bills Payable	1,200	800	Stock	30,000	23,400
Provision for Tax	16,000	18,000	Bills Receivable	2,000	3,200
R.B.D.	400	600	Debtors	18,000	19,000
			Cash at Bank	6,600	15,200
	1,55,600	1,55,800		1,55,600	1,55,800

Additional Information :

(a) Depreciation charged and plant was Rs. 4,000 and Machinery was Rs. 4,000

(b) Provision for Taxation of Rs. 19,000 was made during the year 2011.

(c) Inderium dividend of Rs. 8,000 was paid during the year 2011.

You are required to prepare Fund Flow Statement.

11. What are the types of working capital? What factors determine the working capital requirement of a firm?

12. The following section is the Balance Sheet of a X Company as on 31st March 2011.

Liabilities	Rs.	Assets	Rs.
Share Capital	2,00,000	Land and Building	1,40,000
Profit and Loss A/C	30,000	Plant and Machinery	3,50,000
General Reserve	40,000	Stock	2,00,000
12% Debenture	4,20,000	Sundry Debtors	1,00,000
Sundry Creditors	1,00,000	Bills Receivable	10,000
Bills Payable	50,000	Cash at Bank	40,000
	8,40,000		8,40,000

Calculate :

(a) Current Ratio

(b) Quick Ratio

(c) Inventory to working capital

(d) Debt to Equity Ratio

(e) Proprietary Ratio

(f) Capital Gearing Ratio

(g) Current Assets to Fixed Assets.

13. What budgetary control? How are budgets used for decision making process in management?

14. What are the functions of Financial Management?

15. From the given information prepare cost sheet and calculate direct labour and profit

	Rs.
Direct raw materials	4,20,000
Prime Cost	7,20,000
Factory overheads	2,84,000
Opening work in progress	20,000
Closing work in progress	18,000
Administration Expenses	1,50,000
Opening Stock of finished goods	1,60,000
Closing Stock of finished goods	2,10,000
Selling overheads	1,90,000
Sales	18,00,000

DE–723 **402**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2014.

Fourth Semester

INTERNET PROGRAMMING

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Define WWW. Discuss the various services provided by WWW.

2. Discuss the different levels of access protection in Java.

3. Write a VB Script for finding minimum and maximum number in an array.

4. List out the various objects available in Java Script with example.

5. Explain the table formatting tags in HTML with example.

6. Discuss about Netscape Navigator Pluging.

7. Write the steps to use image and frames in web page.

8. Give a brief note on Dynamic graphics.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Discuss in detail about the web design issues.

10. Describe the complete life cycle of a thread.

11. Write a brief note on looping statement in VB script.

12. Explain the structure of Java Script with example.

13. (a) How anchor tags are used in HTML? Explain.

    (b) Explain the use of Browser and Shockwave.

14. Discuss in detail about VDO live technology.

15. Explain web graphics and animation.

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DE–725 **404**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2014.

Fourth Semester

COMPILER DESIGN

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Explain the process carried out by the compiler.

2. Write short notes on regular expression and transition diagram.

3. What is a parse tree? Give the characteristics of the parse tree.

4. Give the model of predictive parser and explain it briefly.

5. Write short notes on postfix translations.

6. Define errors. Write short notes on lexical phase errors.

7. Write the applications of DAG.

8. Write a note on peephole optimization.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Construct NFA and DFA for the regular expression *(aa*/bb*)*.

10. Define shift reduce parser. Generate the sequence of steps for the input string $id_1 + id_2 * id_3$ using a grammar.

11. Explain the construction of canonical CR parsing table. Give an example.

12. Show the bottom up evaluation of the inherited attributes.

13. Explain three address code, quadruples and triples with an example.

14. Define basic blocks. Explain the DAG construction of basic block with an example.

15. Explain code generation and generate a code sequence for the assignment statement

$X = (A + C) - (C - D) - (C - D)$.

DE–726 **501**

DISTANCE EDUCATION

M.C.A.(S)/M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2014.

Fifth Semester

DISTRIBUTED COMPUTING

(2010 academic year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Define Protocol. Explain Protocol layers and Protocol suites.

2. Discuss the methods available in HTTP.

3. Compare the performance of encryption and secure digest algorithm.

4. Explain how to synchronize the physical clock.

5. Explain Distributed multimedia system.

6. Discuss multicast communication in detail.

7. Explain the distributed object model used for communication between distributed objects.

8. Give an example for implementation of logical clocks by using counters and physical clock.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain various challenges to design distributed systems.

10. Explain in detail

    (a) RMI

    (b) Marshalling

    (c) Sockets

    (d) Request–Reply protocol.

11. Explain public–key cryptography algorithm.

12. Describe Ring-based algorithm and Maekawa's voting algorithm for implementing mutual exclusion in distributed environment.

13. Explain the optimistic concurrency controls in distributed transactions.

14. Discuss the emerging technologies in wireless network.

15. Explain the Concurrency control in Distributed transaction.

DE–727 **502**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2014.

Fifth Semester

WEB TECHNOLOGY

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Write a note on types of CSS.

2. Write a java script to display a welcome button when the form is pressed.

3. Write a note on manifest file in java Beans.

4. How to use bound properties in Java Bean? Explain.

5. Briefly explain the different attributes available in servlets.

6. Write note on Tomcat server.

7. List down the steps involved in JSP MS–access connectivity.

8. Write the steps to connect database in JSP.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. What are the various java script objects? Explain each with an example.

10. Explain the Dom Architecture.

11. (a) Write the features of Bean info interface.

    (b) Explain the steps to create Bean info class.

12. Explain in detail about the life cycle of servlets.

13. Discuss about the MVC setting and JSP environment.

14. How to generate dynamic content pages using JSP? Explain.

15. Write a JSP program to access the student database, based on the student roll no. and display the personal details.

DE–728 **503**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2014.

Fifth Semester

.NET FRAMEWORKS

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Explain the CLS and CTS.
2. Define managed code. List the benefits of managed code.
3. Define assemblies. Explain the shared and private assemblies.
4. Describe the CLR and BCL.
5. Explain the system Object class.
6. Write a note on marshalling and remoting.
7. Explain the dialog box and group box controls in .NET framework.
8. Briefly explain the SOAP.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Discuss the building blocks of .NET framework and its services.
10. Explain the system. String and system Text String Builder classes and its methods.

11. Write a .NET program to explain the concept of multithreading.
12. Describe the attributes and reflection.
13. Give a detailed note on datatable and dataset.
14. Describe the data access using ADO.NET.
15. How do you expose and consume web services using .NET? Explain.

DE–729 **504**

DISTANCE EDUCATION

M.C.A. (S) / M.C.A. (S) (Lateral) DEGREE EXAMINATION, DECEMBER 2014.

Fifth Semester

DATA MINING AND WAREHOUSING

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Discuss the classifications of data mining system.

2. Discuss any five primitives for specifying a data mining task.

3. Explain the difference between the top-down and bottom-up approaches for building data warehouses.

4. Mention the characteristics of data warehouse.

5. List any four key issues to be considered while planning for a data warehouse.

6. Discuss in detail about OLAP operations.

7. What do you mean by Aggregation? Explain.

8. Discuss the metadata and its types.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Discuss the various phases of knowledge discovery process with a neat diagram.

10. What is meant by Association rule mining? Explain the apriori association rule mining with an example.

11. Explain in detail the partition strategy used in data warehouse. What is the need for it?

12. Discuss the operational design of data warehouse.

13. What do you mean by data tuning? Explain.

14. Explain in detail about backup and recovery techniques used in data warehouse.

15. Explain the different techniques available for testing the data in data warehouse.

DE–730 **601**

DISTANCE EDUCATION

M.C.A./M.C.A. (S) (Lateral) DEGREE EXAMINATION, DECEMBER 2014.

Sixth Semester

MOBILE COMMUNICATIONS

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Write short notes on DECT and TETRA.

2. Explain the broadcast system in detail.

3. Write short notes on wireless ATM handover.

4. Give a detailed note on HYPERLAN.

5. Briefly explain about tunneling and encapsulation.

6. List out the disadvantages of mobile TCP.

7. Explain about snooping TCP.

8. Discuss the principles of transport layer security.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Discuss GSM services and architecture in detail.

10. Discuss about digital video broadcasting.

11. Explain WATM services and functions.

12. Give a detailed account of mobile Adhoc network.

13. Explain in detail DHCP and give its usages.

14. Discuss about transaction oriented TCP.

15. Write an overview about wireless session protocol.

DE–731 **602**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2014.

Sixth Semester

MIDDLEWARE TECHNOLOGY

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Describe the key features of NET architecture and list the benefits.

2. Explain about the convergence of CORBA and RMI object models.

3. Explain the concept of session bean and its types in detail.

4. Describe BMP in detail.

5. How do you build database tables in SQL server?

6. Write short note on IIS.

7. Write a database application to display the details of student table in a datagrid control.

8. Explain the functions of WSDL and SOAP in web service.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. (a) Describe in detail about Model View Controller Architecture. (8)

(b) Discuss advantages and disadvantages of client server. (7)

10. (a) Explain the architecture of CORBA in detail. (8)

(b) What are JAVA Servlets? What are the advantages of servlets over CGI? (7)

11. (a) Describe the components of entity bean. (7)

(b) Write in detail about the Container Managed Persistence and explain its responsibilities. (8)

12. Explain about .net architecture with neat sketch.

13. (a) Explain the validation controls in ASP .NET. (7)

(b) Describe the usage and properties of the following ASP .NET controls with suitable examples

(i) AdRotator Controls (4)

(ii) Calendar Controls. (4)

14. Explain in detail about the various Provider and Consumer Objects with suitable examples.

15. Explain in detail about web service architecture.

DE–732 **603**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2014.

Sixth Semester

AGENT BASED INTELLIGENT SYSTEM

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Define problem – solving agent. Differentiate between toy problem and real world problem.

2. Explain constraint satisfaction problem with examples.

3. Write short notes on knowledge based agents.

4. Bring out the significance of unification.

5. Explain state space search.

6. Explain the utility theorem.

7. Give a brief note on Bayes rule and its use.

8. Write short notes on the importance of learning using relevance information.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Give a detailed note on the History of artificial intelligence.

10. Bring out the significance of agents and the nature of environments.

11. Illustrate forward chaining with suitable example.

12. Explain objects, actions and events with suitable diagrams.

13. Describe partial order planning.

14. Explain the role of inference in temporal models.

15. Explain in detail about the different types of statistical learning methods.

DE–733 **604**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2014.

Sixth Semester

OPEN SOURCE ARCHITECTURE

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Explain briefly the following commands Ls, Cp, more, find.

2. How to configure X windows? Explain.

3. Explain the process of SAMBA installation and configuration in detail.

4. What are the steps to follow setting up the SQUID proxy server?

5. Briefly explain system logging.

6. Explain the syntax of make files.

7. Write the python program to calculate the total, average and grade of student mark.

8. Write a GKT program which constraints a button and when we press the same it display "WELCOME".

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. What is UNIX account and how to configure, create, test and remove a user account.

10. Explain the two stages of boot up process in linux OS.

11. Explain the detailed analysis of compilation process.

12. Explain the versioning and management tools – RCS.

13. Explain the commands which can be used at Gdb prompt in GNU debugger with an examples.

14. Write the python program to get 5 employee details calculate the employee HRA 12%, DA 10%, other allowance 15% from basic and the net salary.

15. Explain the X windows server architecture in detail.

DE–8850 **105**

DISTANCE EDUCATION

M.C.A. (S) DEGREE EXAMINATION, DECEMBER 2014.

First Semester

DATA STRUCTURE AND USING C LAB

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select ONE question by the LOT system and give it to the student.

1. (a) Write a program to sort the given a set of numbers using Heap sort.

   (b) Write a 'C' program to find the variance and standard deviation of 'n' numbers.

---------- Cut here ----------

2. (a) Write a program to search a number from the given set of numbers using binary search.

   (b) Write a C program to find the factorial of a given number using function.

---------- Cut here ----------

3. (a) Write a program to convert the infix expression to postfix expression.

   (b) Write a function called 'prime' that returns 1 if its argument is a prime number and returns zero otherwise.

4. (a) Create a Queue and add elements into the queue and delete elements from the same queue.

(b) Write a program to solve the given quadratic equation $ax^2 + bx + c = 0$.

---------- Cut here ----------

5. (a) Write a program to generate a calculator, which should perform the following functions.

(i) Addition

(ii) Subtraction

(iii) Multiplication

(iv) Division

Input any two integers and a character for operation (+ for addition, – for subtraction, * for multiplication, /for division.)

(b) Write a program to implement the linked stack.

DE–8851 **106**

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, DECEMBER 2014.

First Semester

RELATIONAL DATABASE MANAGEMENT SYSTEMS LAB

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select ONE question by the lot system and give it to the student.

1. Create the following two tables.

   College – info; College – Code, College – name, address Faculty – info: College – Code, Faculty – Code, Faculty – name, qualification, experience, address.

   Write query to do the following :

   (a) List all those faculty members whose experience is greater than or equal to 10 years and have M.Tech degree.

   (b) List all those faculty members, who have at least 10 years of experience but do not have M.Tech degree.

   (c) Display all those records from faculty–info table where name starts with 'a' or experience not less than 10 years.

---------- Cut here ----------

2. Create a table to represent customer support system with the following fields.

   Customer–id, Customer–name, address, phone–no, date, complaint, feedback

   (a) Create the above table by properly specifying the primary keys and foreign keys.

   (b) List the customers who registered complaint for same product.

   (c) List the customers who registered complaint in a day.

   (d) List the customers who are unsatisfied.

3. Create a table of employee (emp–number, name, dept, salary, designation) and department (dept number, dept name).
Write query to do the following.
   (a) List the employees who is getting salary greater than 10,000
   (b) List the employees who belong to IT department and getting a salary below 15,000
   (c) Find out the average salary of the employees of manufacturing department.

---------- Cut here ----------

4. Consider the following tables of a motor cycle company
BRANCH (branch name, address, sales, date)
SALES (branch – id, branch name, category, date)
MODEL (model – id, model name, price)
   (a) Prepare the highest sales of a particular model in a day.
   (b) List the branch name and its address where the sales category is poor.
   (c) List the average sales done by every branch in a month.
   (d) List the average sales of a model in all branches.

---------- Cut here ----------

5. Create a tables of a mark list contains the following.
Roll no, Name, Mark 1, Mark 2, Mark 3, Total, Grade using the MARKS table do the following :
   (a) Search a Roll No and display total marks and grade.
   (b) For passing a subject, minimum, mark is 40. For the failed students grade is "F". Fort the passed students grade is set as
   "A' if total > 240
   'B' if total between 180 and 240
   'C' if total between 150 and 180
   'D' otherwise exception must be raised if Roll No could not find.

6. Create the following tables with the constraints specified.

Book (book – id, book name, author – id, publisher – id, pub – date, category and price)

AUTHOR (author – id, name, city, state)

PUBLISHER (pub – id, name, city, state)

Write SQL to do the following.

(a) Display the books in the category 'Computer Science'

(b) List the count of books in each category

(c) List the books published after 2006

(d) List the names of book having highest price

(e) List the average price of each category of the books

(f) Display the medical books having, price greater, than the price of the category 'Computer Science'.

---------- Cut here ----------

7. Consider the following data base for an advertising company

Advertisement (type, time, product)

Agreement period (6 months, 1 year, 2 years, above)

Category (id, rating)

Feedback (30, 50, 80, above)

Write SQL statements to do the following.

(a) Display the highest feedback of a visual advertisement

(b) Display the advertisement which is having long time display

(c) Display the customers who is holding the rating above 90% and agreement above 2 years.

8. Create a table to represent theatre ticket reservation consisting of the following fields customer name, sex, date, seat number, show time, class.
   (a) Create the above table by properly specifying primary keys and foreign keys
   (b) Find the customer who have more than 3 tickets
   (c) Display all the female customers in a show
   (d) Display the shows which is houseful in a day.

---------- Cut here ----------

9. Create a database of a employee table with the following fields.
   basic pay, DA, MA, HRA, CCA, PF, Loan, Net pay, gross pay
   DA = 70% of basic pay, MA = 1000Rs, HRA = 25% of Basic pay, PF = 15% of basic pay, CCA = Rs. 1500,
   Gross pay = basic pay + DA + MA + HRA + CCA
   Net pay = gross pay – (PF + Loan)
   Prepare the salary statement of each employee.

---------- Cut here ----------

10. A online shopping portal displays the list of books, CD's, DVD's, pen drives, peripherals of various companys. They sell each goods with a profit of 10% on each sale. Write SQL statement to find the following.
   (a) List the items sold for a particular brand
   (b) List the maximum sales of an item in a day
   (c) Display the profit earned at end of the day
   (d) List the item which is not at all sold in a month.

DE–8852 **205**

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, DECEMBER 2014.

Second Semester

C++ AND GRAPHICS LAB

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

Examiner has to choose ONE question for each candidate by lot system.

Each subdivision carries 50 marks.

1. (a) Write a C++ program to draw a straight line using Breshnham's algorithm.

   (b) Develop a C++ program to perform transformations (scaling, rotation and translation on a two dimensional square)

---------- Cut here ----------

2. (a) Develop a C++ program to draw a circle divide it into four quadrants and fill up different colors in each quadrant.

   (b) Write a C++ program to draw a straight line using simple DDA algorithm.

---------- Cut here ----------

3. (a) Develop a C++ program to draw a square within a circle such that the four edges of the square touches the circle.

   (b) Write a C++ program to draw non-linear objects arcs and curves.

4. (a) Write a C++ program to manipulate the graphical object using SHADING technique.

   (b) Develop a C++ program to draw a straight line using Breshnham's algorithm.

---------- Cut here ----------

5. (a) Develop a C++ program to draw an ellipse using Breshnham's algorithm.

   (b) Write a C++ program to perform scaling transformation on two dimensional object.

---------- Cut here ----------

6. (a) Write a C++ program to implement polygon dipping algorithm.

   (b) Develop a C++ program to draw a circle and fill it with any color.

---------- Cut here ----------

7. (a) Write a C++ program to draw a ball and perform animation on it.

   (b) Develop a C++ program to draw a straight line using Breshnhams algorithm.

---------- Cut here ----------

8. (a) Write a C++ program to show graphical transformation on three dimensional objects.

   (b) Develop a C++ program to draw a circle using Breshnhams algorithm.

9. (a) Write a C++ program to draw non-linear object circle and three concentric circle.

   (b) Develop a C++ program to draw a cube.

---------- Cut here ----------

10. (a) Write a C++ program to draw Olympic symbol using circles.

    (b) Develop a C++ program to draw a circle and fill it with any color.

DE–8853 **206**

DISTANCE EDUCATION

M.C.A. (S) DEGREE EXAMINATION, DECEMBER 2014.

Second Semester

UNIX AND SHELL PROGRAMMING LAB

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

Record Note Book : 5
Algorithm/Flow chart : 15
Program : 35
Execution : 35
Result : 10
Total : 100

Each candidate should select ONE question using Lot system.

Each subdivision carries 50 marks.

1. (a) Write a shell program to move a file in one directory to another directory.

   (b) Write a shell program to find how many weeks remaining in a current year.

2. (a) Write a menu driven shell program for the following:

   (i) To see if a user already exists.

   (ii) Enter any five string on an existing file.

   (b) Write a shell program to accept two numbers and find the sum.

---------- Cut here ----------

3. (a) Write a shell program to accept a set of names and display them in alphabetical order.

   (b) Write a shell program to compare two strings and determine if they are the same.

---------- Cut here ----------

4. (a) Write a shell program to perform the following:

   (i) To delete a directory.

   (ii) To list the files in a directory.

   (b) Write a shell program to check if the character entered is a vowel or not.

5. (a) Write a shell program for the following:

   (i) Enter fruit names in a files.

   (ii) Sort the names in existing file.

   (b) Write a shell program to display the following output.

   1

   1 2

   1 2 3

   1 2 3 4

   1 2 3 4 5

---------- Cut here ----------

6. (a) Write a shell program to accept 8 numbers and print the sum of all positive numbered entered.

   (b) Write a shell program that checks whether an input character is a digit or not.

---------- Cut here ----------

7. (a) Write a shell program to find the value of $\sin x$, whose value is given by $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!}..$ upto 10 terms.

   (b) Write a shell program to accpet a number from 1 to 12 and display the corresponding month (1 → January, 2 → February, 12 → December)

DE–8854 **305**

DISTANCE EDUCATION

M.C.A. (S)/ M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2014.

Third Semester

NETWORK LAB

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

Break-up marks

Record Note Book	: 5
Algorithm/Flow chart	: 15
Program	: 35
Debugging and Execution	: 35
Result	: 10
Total	: 100

Examiner has to choose any ONE question for each candidates by LOT system.

Each subdivision carries 50 marks.

1. (a) Write a Java Program to Encrypt the word "Hello".

   (b) Write a Java Program to find factorial of given number using RMI client, server technique.

2. (a) Write a Java program to set up a chat server and chat client.

   (b) Write a Java program to read a file from the remote system.

---------- Cut here ----------

3. (a) Write a Java program to send message from one system to another.

   (b) Write a Java program to setup echo server and echo client.

---------- Cut here ----------

4. (a) Write a Java program to find factorial of given number using factorial interface technique.

   (b) Write a Java program to display the users and their IP addresses logged in the network.

---------- Cut here ----------

5. (a) Write a Java program to add two numbers using RMI, client, server technique.

   (b) Write a Java program to setup time server and time client.

---------- Cut here ----------

6. (a) Write a Java program to read a file from the remote system.

   (b) Write a Java program to Encrypt given word.

7. (a) Write a Java program to setup echo server and echo client.

   (b) Write a Java program to find factorial of given number using RMI client, server technique.

---------- Cut here ----------

8. (a) Write a Java program to send message from one system to another.

   (b) Write a Java program to display the users and their IP addresses logged in the network.

DE–8855 **306**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2014.

Third Semester

VISUAL PROGRAMMING LAB

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

Break-up marks

Record Note Book : 5
Algorithm/Flow chart : 15
Program : 35
Debugging and Execution : 35
Result : 10
Total : 100

Examiner has to select ONE question and give it to the students by the lot system.

1. (a) Write a VB program to perform the function procedure.

   (i) To find the factorial of the given number.

   (ii) To convert the temperature in Fahrenheit into celsius.

   (b) Write a visual C++ program to get the status of the shift and toggle keys using MFC.

2. (a) Write a VB program using ActiveX control to create a textbox that accepts numeric value. It includes back color, forecolor and text.

   (b) Write a visual C++ program to create a window using C++ source files.

---------- Cut here ----------

3. (a) Write a VB program to enter a string and check, if it is a palindrome or not.

   (b) Write a visual C++ program for keyboard events.

---------- Cut here ----------

4. (a) Write a VB program to enter a file name in a textbox and search for the given file name in your system.

   (b) Write a visual C++ program to create a menu and toolbar.

---------- Cut here ----------

5. (a) Write a program in VB to create an account in a bank along with the fields AccNo, NAME, SEX, ADDRESS, AMOUNT and write a procedure to deposit and withdraw a sum of amount to/from the SB account.

   (b) Write a visual C++ program to create document – view – frame – menu using Appwizard.

DE–8856 **405**

DISTANCE EDUCATION

M.C.A.(S)/M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2014.

Fourth Semester

INTERNET PROGRAMMING LAB

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

Break-up of marks

Record Note Book : 5
Algorithm/Flow chart : 15
Program : 35
Debugging and Execution : 35
Result : 10
Total : 100

Examiner should select and give ONE question to each candidate by Lot system.

1. (a) Create a webpage to display student details that contains S.No., Register number, name of the student, fathers name, address, phone number etc. using tables tag.

   (b) Write an applet to display a string in blue color against a green background. Then reset the original background and foreground colors and display the string again.

2. (a) Create a login page with user name and password from the user and validates them. If the user name is "root" and the password is "tiger" then the user is authorized otherwise reject using VB script

(b) Design a web page for the following

(i) Set an image as a link

(ii) Open a link in a new browser window

(iii) Jump to another part of a document (on the same page)

(iv) How to link to a mail message with CC, BCC and Subject Entries

(v) Redirect a user to another URL after 10 seconds.

---------- Cut here ----------

3. (a) Write a Java program to display an Employee Detail available in a database.

(b) Design a web page to display University Mark sheet using table tag.

4. (a) Write a VB script program to compute bill for Internet Browsing. The conditions are given below

Hours	Rupees
0.5	10
1	15

(b) Create a web page having two frames one frame containing the two links (Flowers and Animals) and another frame should display the detail of the link (Flowers and Animals). When link is clicked appropriate contents should be displayed on Frame 2.

---------- Cut here ----------

5. (a) Write a java program that lets the user to create Pie charts. Design your own user interface with Swings.

(b) Develop a web page for Job recruitment agency to an IT Industry.

DE–8857 **406**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (LAT) DEGREE EXAMINATION, DECEMBER 2014.

Fourth Semester

LAB X – COMPILER DESIGN

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

Record Note Book : 5
Algorithm/Flow chart : 15
Program : 35
Debugging and Execute : 35
Result : 10
Total : 100

Examiner should select and give ONE question to each candidate by LOT system.

1. (a) Write a C program to recognize constant in the given source code.

   (b) Write a C program for DFA to accept the regular expression $(a/b)^* abb$.

---------- Cut here ----------

2. (a) Write a C program to check the production rule whether it is an operator grammar or not.

   $S \rightarrow aBcd$

   $S \rightarrow aBdc$.

   (b) Write a C program to produce line numbers in the given source code.

3. (a) Write a C program to construct shift reduce parser for the input string 'cdcd' using the grammar.

$S \to CC$, $C \to CC$, $C \to d$.

(b) Write a C program to draw the transition diagram for DFA by accepting regular expression $(0/1)^*1$.

---------- Cut here ----------

4. (a) Write a C program to convert given infix $a + b * c - d$ into postfix.

(b) Write a C program to construct predictive parsing table for the production rules.

$A \to i\,B\,t\,A$, $A \to a$, $B \to b$.

---------- Cut here ----------

5. (a) Write a C program to accept a string and check whether it satisfies the following production rules using operator precedence grammar.

$S \to AbB$, $A \to aBb$, $B \to a / Bba$

(b) Write a C program to recognize keywords in the given source code.

6. (a) Write a C program to construct right most derivation to derive the word $(a,(a,a))$ using the grammar.

$S \to a/(T),\ T \to T,S/S$

(b) Write a C program to translate the expression $(a+b)*(b-c)$ into postfix notation and three address code.

---------- Cut here ----------

7. (a) Write a C program to construct a SLR parsing table for the following grammar $E \to E+T/T$, $T \to T*F/F$, $F \to (E)/id$.

(b) Write a C program to recognize identifier in the given source code.

---------- Cut here ----------

8. (a) Write a C program to find the LEADING set for the grammar.

$S \to a,\ S \to (T),\ T \to T,S,\ T \to S$

(b) Write a C program to construct LR parsing table for the grammar.

$A \to BB,\ B \to Bb/c$.

DE–8858 **505**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2014.

Fifth Semester

.NET LAB

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

Examiner should select and give ONE question to each candidate by lot system.

Each subdivision carries 50 marks.

1. (a) Create a student table and insert few records using ADO.NET.

   (b) Create an ASP.NET application to view the first, last, move next and move previous records of student table.

---------- Cut here ----------

2. (a) Create an application to insert, delete and update the details of employees.

   (b) Display the salary details of all the employees using the Gridview control.

3. (a) Write an ASP.NET application to create a web page for your university.

(b) Write an ASP.NET application to create a staff bio-data.

---------- Cut here ----------

4. (a) Develop a splash screen to show a message to the passengers regarding the train number, destination, arrival time, departure time and platform number for every 5 minutes.

(b) Create an ASP.NET application to validate the name and pin code field of a customer form.

---------- Cut here ----------

5. (a) Create a customer table and insert few records using ADO.NET.

(b) Generate a crystal report for the customers buying a soap twice in a month.

---------- Cut here ----------

6. (a) Create a simple text editor.

(b) Write a program to upload and download a file into e-mail using ASP.NET.

DE–8859 **506**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2014.

Fifth Semester

WEB TECHNOLOGY LAB

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

Examiner should select and give ONE question to each candidate by LOT system.

1. (a) Write a JSP program for employee pay bill system.

   (b) Write a servlet program to create a chat application.

---------- Cut here ----------

2. (a) Write an XML document to display current system IP Address, System Id, System name, port number. Write an XSL style sheet and attach that to the XML document.

   (b) Write an ASP program to vaildate the username and password using Javascript.

---------- Cut here ----------

3. (a) Write a Javascript to animate an image.

   (b) Write a servlet program to shown "Your University Name".

4. (a) Create a Home page for your department information using web graphics.

   (b) Write an ASP program to prepare Telephone bill using Javascript.

---------- Cut here ----------

5. (a) Write a servlet program for mail services.

   (b) Create a web page for your exam time table using HTML table.

---------- Cut here ----------

6. (a) Write a JSP program to show the full detail history of patient based on the patient id using JDBC.

   (b) Write a servlet program to display the request and response header information.

---------- Cut here ----------

7. (a) Write an ASP programs for students Rank list in your class using Javascript.

   (b) Write a servlet program to show your system IP address date and time.

---------- Cut here ----------

8. (a) Create a web page for your college offered courses using web graphics and text linus.

   (b) Write a servlet program to display your server information.

DE–8860 **605**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2014.

Sixth Semester

OPEN SOURCE PROGRAMMING LAB

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select ONE question by the lot system and give it to the student.

1. (a) Write a PHP program to find GROSS. SALARY for the given basic salary (bs) of employee using following condition :

If bs > 8000, da = 95% of bs and hra = 11% bs

If bs<7999 and more than 5000, da = 90% of bs and hra = 15% bs

Else da = 98% of bs and hra = 1000

GS = bs + da + hra

(b) Write a PHP program for the following conversion

(i) binary to decimal

(ii) count all hidden files in a current working directory.

2. (a) Write a PHP program to find "Mod" calculation.

   (b) Write a PHP program for the following :

      (i) To sort an employee based on salary and print it in a reverse order.

      (ii) To print how many project-managers are present in a database.

      (iii) To print the records other than the project-manager.

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3. (a) Write a PHP program to check whether given number is even or odd using positional parameters.

   (b) Write a PHP program to find leap year calculation.

---------- Cut here ----------

4. (a) Write a shell script to print terminal number of first seven logged user in system.

   (b) Write a shell script for the following conversion.

      (i) binary to hexadecimal

      (ii) binary to octal.

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5. (a) Write a server side PHP program that display EMPNO, EMPNAME, NETSALARY of a employee in tabular format by accepting user inputs from a HTML form.

   (b) Create a my SQL table and execute queires to read, add, remove and modify a record form the table.

DE–8861 **606**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2014.

Sixth Semester

MULTIMEDIA TOOLS LAB

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

Break-up of marks :

Record Note Book	5
Algorithm/Flow chart	15
Program	35
Debugging and Execution	35
Result	10
Total	100

Examiner has to select ONE question by the LOT system and give it to the student.

1. (a) Develop an animation for butterfly with three flowers in the garden.

   (b) Implement a program to create a Maze using Maya.

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2. (a) Implement a program in flash for getting rainfall effect.

   (b) Implement a program to create study room using rendering.

3. (a) Develop an animation using button in flash for making five faces.

   (b) Implement an animation for eye protection awareness.

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4. (a) Develop a webpage using dream viewer for paper competition.

   (b) Implement a program to create a Maze using Maya.

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5. (a) Implement a program in flash for bouncing ball with a pad.

   (b) Implement a program to create study room using rendering.